

Project Report on: CelebAMask Face Parsing

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1 Methodology

1.1 Model Architecture

The model employed in this project adopts the Efficient Bottleneck Unit (EBU)[5] architecture, a lightweight encoder-decoder framework originally designed for urban street scene segmentation tasks. Compared to the original EBUNet[5], which limits feature extraction and fusion to 1/4 and 1/8 scales, an auxiliary 1/2 resolution scale was integrated in this work to improve the prediction accuracy of edge pixels.

1.1.1 Core Components

The core architecture of EBUNet comprises three primary components: the Efficient Bottleneck Unit (EBU) for hierarchical feature extraction, the Image Partition Attention Module (IPAM) for feature map refinement, and the Lightweight Attentional Decoder (LAD) for multi-scale feature fusion. Figures 1–3 show the detailed structures of these modules.

The EBU employs parallel depth-wise standard convolution and depth-wise dilated convolution. This design effectively reduces the number of parameters, while allowing the network to capture multi-scale contextual information by varying the dilation rates of the depth-wise dilated convolutions. To compensate for the limited feature representation caused by depth-wise convolutions, lightweight attention mechanisms are integrated into the IPAM and LAD modules. Specifically, in the LAD module, to effectively recover information, a Channel Attention Module (CAM)[5] is applied to 1/8-scale feature maps to focus on global context, while a Spatial Attention Module (SAM)[5] is applied to 1/4-scale feature maps to refine boundary details. To further enhance the model’s learning capability for boundaries, 1/2-scale features are also preserved and concatenated into the final feature map.

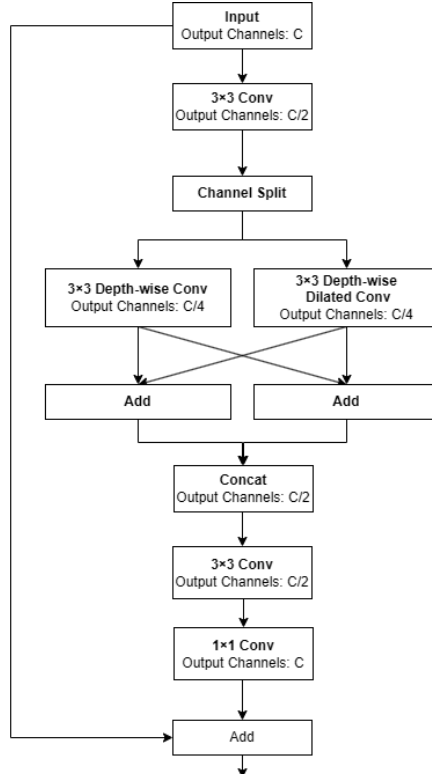


Figure 1: EBU Architecture

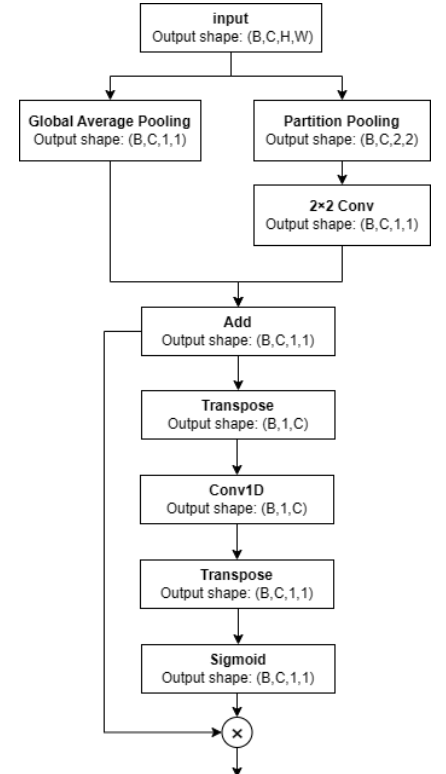


Figure 2: IPAM Architecture

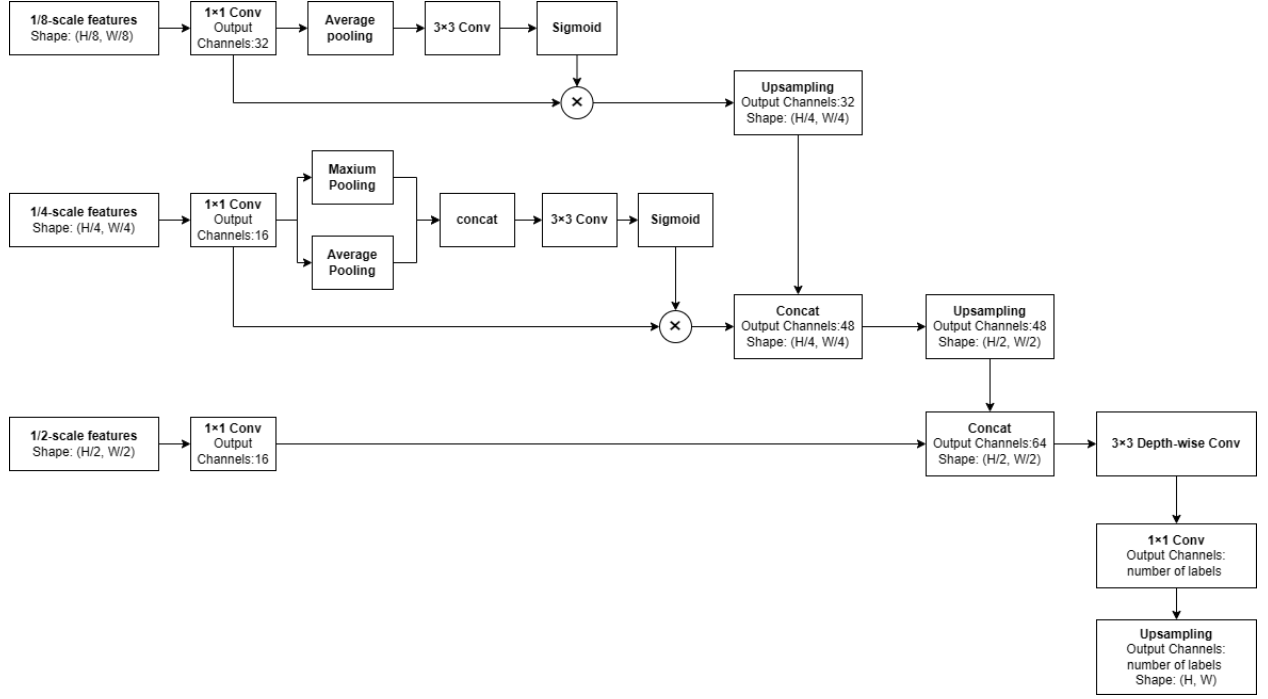


Figure 3: LAD Architecture

1.1.2 Overall Architecture

The overall architecture of the final model is illustrated in Figure 4. The input image first passes through an Initial Unit consisting of three consecutive convolutional layers to eliminate redundant information and extract basic contextual information. Subsequently, the output is concatenated with the three-channel average pooling result of the original image before being fed into the EBU Block. The output of the EBU Block is then further optimized by the IPAM module. Finally, the original pooling result, the EBU block output, and the IPAM-optimized features are concatenated to form the feature maps to be input into the LAD.

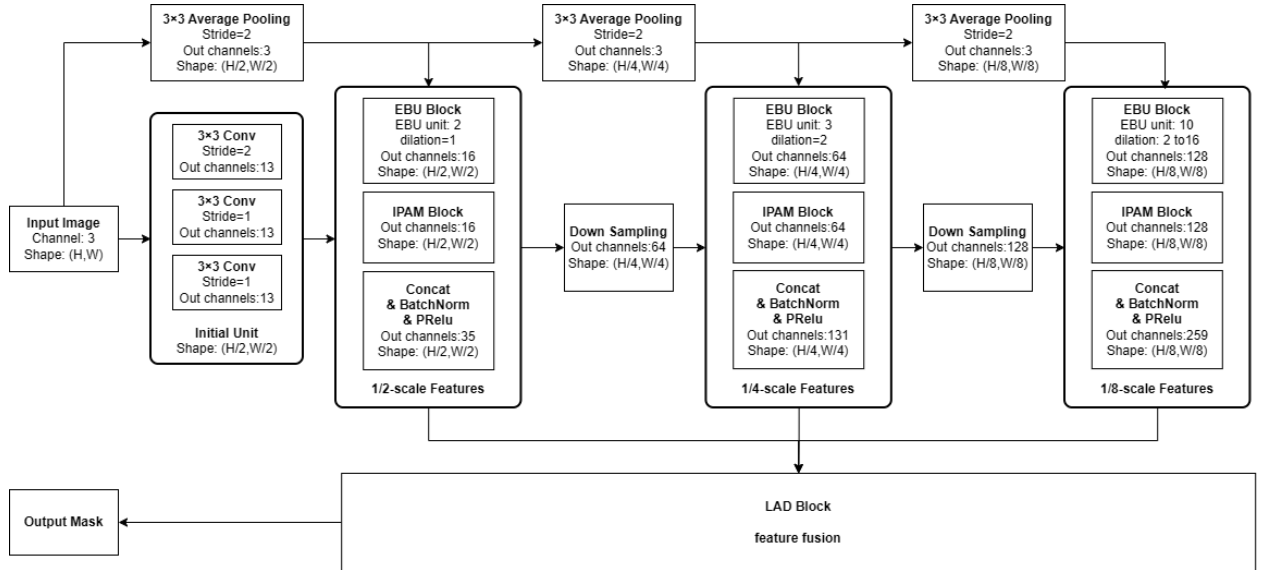


Figure 4: Overall Architecture

The aforementioned feature extraction process is performed three times, each accompanied by a down-sampling operation. These stages yield feature maps with resolutions of 1/2-scale (35 channels), 1/4-scale (131 channels), and 1/8-scale (259 channels), respectively. In the EBU Block, both the number of EBU units

and their dilation rates are gradually increased: the first stage consists of 2 EBU units using only standard depth-wise convolutions; the second stage contains 3 EBU units with a dilation rate of 2; and the third stage employs 10 EBU units with dilation rates ranging from 2 to 16. Finally, the features extracted from the three stages are further fused within the LAD, followed by the final label prediction.

1.1.3 Number of Parameters

The parameter count of the final model is 1,536,362, accounting for 84.3% of the maximum allowable parameters.

1.2 Optimization Strategy

1.2.1 Loss Function

The loss function was defined as the linear combination of weighted Cross-Entropy loss and Soft-Dice loss[1], which can be formulated as Equation (1):

$$\mathcal{L}_{total} = \alpha \cdot \mathcal{L}_{WCE} + \beta \cdot \mathcal{L}_{Dice} \quad (1)$$

The weighted cross entropy loss in Equation (1) can be expressed as

$$\mathcal{L}_{WCE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C w_c \cdot \mathbb{I}(y_i = c) \log(P(y_i = c|x_i)) \quad (2)$$

To address the class imbalance problem, the Median Frequency Balancing strategy was employed to calculate the class weights w_c . The weight for each class c is determined by the ratio of the median of all non-zero class frequencies to the specific frequency of class c . To prevent gradient explosion, the calculated weights are clipped within the range $[0.5, 4.0]$.

The Soft-Dice loss in Equation (1) can be expressed as

$$\mathcal{L}_{Dice} = 1 - \frac{1}{N} \sum_{c=1}^N \frac{2 \sum_{i=1}^M p_{i,c} g_{i,c} + \epsilon}{\sum_{i=1}^M p_{i,c} + \sum_{i=1}^M g_{i,c} + \epsilon} \quad (3)$$

where N denotes the number of semantic classes, M is the total number of pixels in an image, $p_{i,c}$ represents the predicted probability of pixel i belonging to class c , $g_{i,c}$ is the ground truth in one-hot encoding, and ϵ is a smoothing factor to avoid division by zero.

Incorporating Soft-Dice loss primarily aims to optimize regional overlap, providing a robust solution to severe class imbalance while refining the boundaries of small objects.

1.2.2 Two-stage Optimization Protocol

A two-stage optimization strategy was employed in this project to enhance the network’s capability in learning boundaries and hard examples. During the initial 30% of the training iterations, the loss function was calculated following a standard protocol, which can be formulated as:

$$\mathcal{L}_{total} = 1.0 \cdot \mathcal{L}_{WCE} + 0.1 \cdot \mathcal{L}_{Dice} \quad (4)$$

After the model has sufficiently converged on majority categories, the OHEM[6] strategy was applied to further refine its performance, the total loss can be formulated as Equation (5) to Equation (7)

$$\mathcal{L}_{total} = 1.0 \cdot \mathcal{L}_{OHEM} + 0.5 \cdot \mathcal{L}_{Dice} \quad (5)$$

$$\mathcal{L}_{OHEM} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \ell_i \quad (6)$$

$$\mathcal{M} = \begin{cases} \{i \in \mathcal{P} \mid P(y_i|x_i) < \tau\} & \text{if } \text{count}(P(y_i|x_i) < \tau) \geq K \\ \text{top-}K\{i \in \mathcal{P} \mid \ell_i\} & \text{otherwise} \end{cases} \quad (7)$$

where ℓ_i is the weighted Cross-Entropy loss derived from Equation (2), $\mathcal{P}$ is the set of all valid pixels, τ is the confidence threshold, and K is the minimum number of pixels to be kept for back-propagation.

1.3 Evaluation Metrics

During the training and evaluation phases, the model’s final performance was assessed using per-class F_1 scores and the macro-average F_1 score. The calculation can be formulated as Equation (8) to Equation (11):

$$\text{Precision}^{(c)} = \frac{TP^{(c)}}{TP^{(c)} + FP^{(c)} + \epsilon} \quad (8)$$

$$\text{Recall}^{(c)} = \frac{TP^{(c)}}{TP^{(c)} + FN^{(c)} + \epsilon} \quad (9)$$

$$F_1^{(c)} = \frac{2 \cdot \text{Precision}^{(c)} \cdot \text{Recall}^{(c)}}{\text{Precision}^{(c)} + \text{Recall}^{(c)} + \epsilon} \quad (10)$$

$$\text{Macro-}F_1 = \frac{1}{C} \sum_{c=0}^{C-1} F_1^{(c)} \quad (11)$$

2 Experiments

2.1 Dataset and Preprocessing

Given the limited dataset consisting of 1,000 images from CelebAMask-HQ[3], the data were partitioned into training and validation sets at a 9:1 ratio. To enhance generalization performance, several data augmentation techniques were applied to the training set, including random horizontal flipping, small-angle random rotation, color jittering, and CutMix. It is worth noting that during horizontal flipping, labels distinguishing left and right sides in the masks were manually swapped to maintain anatomical consistency. Furthermore, all interpolation operations for masks utilized the nearest-neighbor method to preserve label integrity.

2.2 Training Settings

Model training was conducted on a single NVIDIA GeForce RTX 4070 GPU. Network parameters were initialized using the Kaiming initialization method[2]. During training, the AdamW optimizer[4] was employed with a batch size of 8, an initial learning rate of 5×10^{-4} , and a weight decay of 1×10^{-5} . A cosine annealing scheduler was applied throughout the process to adjust the learning rate. The entire training procedure consisted of 20,000 iterations, taking approximately 4 hours to complete.

2.3 Evaluation Results

The model achieved a final F-measure score of 0.843 on the test set.

3 Experimental Analysis

The primary challenges encountered in this experiment stemmed from the scarcity of training data and the poor recognition of low-frequency, small-scale labels (e.g., earrings and necklaces). Findings during the

development phase suggested that when training data is extremely limited, the returns from simply increasing model depth or parameter count were lower than those from optimizing data augmentation strategies. Although weighted Cross-Entropy loss could facilitate the learning of infrequent categories, excessively high weights often led to convergence instability. Furthermore, while the OHEM method was shown to improve the Macro- F_1 score by approximately 0.03, a category-wise analysis revealed that its contribution primarily lay in refining boundary pixel representations rather than enhancing the recognition of rare semantic labels.

Moreover, the over-sampling strategy was also evaluated for low-frequency categories during the training process. However, this approach resulted in a noticeable decline in F-measure scores on the test set, suggesting that the over-fitting it induced outweighed its potential benefits. Consequently, the over-sampling method was deprecated in the final training pipeline.

4 Acknowledgments

4.1 Codebases

The implementation of this project is based on the following codebases:

CelebAMask-HQ (<https://github.com/switchablenorms/CelebAMask-HQ>).

EBUNet (<https://github.com/Skybird1101/EBUNet>).

4.2 Declaration on the Use of AI

The report was polished with the assistance of Google Gemini 3 for language refinement.

References

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